SAMPLE

Time

From Sundials to Black Holes

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Introduction

Time is the continuous progression of existence that occurs in an apparently irreversible succession from the past, through the present, and into the future. Time dictates all forms of action, age, and causality, being a component quantity of various measurements used to sequence events, to compare the duration of events or the intervals between them, and to quantify rates of change of quantities in material reality or in conscious experience.

Time is often referred to as the fourth dimension and the temporal dimension, in addition to the three spatial dimensions. Time is primarily measured in linear spans or periods, ordered from shortest to longest. Practical, human-scale measurements of time are performed using clocks and calendars, reflecting a 24-hour day collected into a 365-day year linked to the astronomical motion of the Earth.

Scientific measurements of time instead vary from Planck time at the shortest to billions of years at the longest.

Measurable time is believed to have effectively begun with the Big Bang 13.8 billion years ago, encompassed by the chronology of the universe. Modern physics understands time to be inextricable from space within the concept of spacetime described by general relativity.

Time can therefore be dilated by velocity and matter to pass faster or slower for an external observer, though this is considered negligible outside of extreme conditions, namely relativistic speeds or the gravitational pulls of black holes.

Throughout history, time has been an important subject of study in religion, philosophy, and science. Temporal measurement has occupied scientists and technologists, and has been a prime motivation in navigation and astronomy. Time is also of significant social importance, having economic value ("time is money") as well as personal value, due to an awareness of the limited time in each day ("carpe diem") and in human life spans.

The concept of time can be complex.

Multiple notions exist, and defining time in a manner applicable to all fields without circularity has consistently eluded scholars. Nevertheless, diverse fields such as business, industry, sports, the sciences, and the performing arts all incorporate some notion of time into their respective measuring systems.

Traditional definitions of time involved the observation of periodic motion such as the apparent motion of the sun across the sky, the phases of the moon, and

the passage of a free-swinging pendulum.

A day is usually 24 hours or 86,400 seconds in length; however, the duration of a calendar day can vary due to daylight saving time and leap seconds.

Isaac Newton subscribed to this realist view, and hence it is sometimes referred to as Newtonian time.

Chapter 1: The Discovery of Time

More modern systems include the Global Positioning System, other satellite systems, Coordinated Universal Time and mean solar time. Although these systems differ from one another, with careful measurements they can be synchronized. In physics, time is a fundamental concept to define other quantities, such as velocity.

To avoid a circular definition, time in physics is operationally defined as "what a clock reads", specifically a count of repeating events such as the SI second.

Although this aids in practical measurements, it does not address the essence of time. Physicists developed the concept of the spacetime continuum, where events are assigned four coordinates: three for space and one for time. Events like particle collisions, supernovas, or rocket launches have coordinates that may vary for different observers, making concepts like "now" and "here" relative.

In general relativity, these coordinates do not directly correspond to the causal structure of events.

Instead, the spacetime interval is calculated and classified as either space- like or time-like, depending on whether an observer exists that would say the events are separated by space or by time. Since the time required for light to travel a specific distance is the same for all observers—a fact first publicly demonstrated by the Michelson-Morley experiment—all observers will consistently agree on this definition of time as a causal relation.

General relativity does not address the nature of time for extremely small intervals where quantum mechanics holds. In quantum mechanics, time is treated as a universal and absolute parameter, differing from general relativity's notion of independent clocks. The problem of time consists of reconciling these two theories. As of 2026, there is no generally accepted theory of quantum general relativity. Methods of temporal measurement, or chronometry, generally take two forms.

The first is a calendar, a mathematical tool for organising intervals of time on

Earth, consulted for periods longer than a day. The second is a clock, a physical mechanism that indicates the passage of time, consulted for periods less than a day. The combined measurement marks a specific moment in time from a reference point, or epoch.

Time is one of the seven fundamental physical quantities in both the International System of Units (SI) and International System of Quantities.

The SI base unit of time is the second, which is defined by measuring the electronic transition frequency of caesium atoms. Artifacts from the Paleolithic suggest that the moon was being used to reckon time by at least 6,000 years ago. Lunar calendars were among the first to appear, with years of either 12 or

13 lunar months (either 354 or 384 days).

Without intercalation to add days or months to some years, seasons quickly drift in a calendar based solely on twelve lunar months.

Lunisolar calendars have a thirteenth month added to some years to make up for the difference between a full year (now known to be about 365.24 days) and a year of just twelve lunar months. The numbers twelve and thirteen came to feature prominently in many cultures, at least partly due to this relationship of months to years.

Other early forms of calendars originated in Mesoamerica, particularly in ancient Mayan civilization, in which they developed the Maya calendar, consisting of multiple interrelated calendars.

These calendars were religiously and astronomically based; the Haab' calendar has 18 months in a year and 20 days in a month, plus five epagomenal days at the end of the year. In conjunction, the Maya also used a 260-day sacred calendar called the Tzolk'in. The reforms of Julius Caesar in 45 BC put the Roman world on a solar calendar.

This Julian calendar was faulty in that its intercalation still allowed the astronomical solstices and equinoxes to advance against it by about 11 minutes per year. Pope Gregory XIII introduced a correction in 1582; the Gregorian calendar was only slowly adopted by different nations over a period of centuries, but it is now by far the most commonly used calendar around the world.

During the French Revolution, a new clock and calendar were invented as part of the dechristianization of France and to create a more rational system in order to replace the Gregorian calendar. The French Republican Calendar's days consisted of ten hours of a hundred minutes of a hundred seconds, which marked a deviation from the base 12 (duodecimal) system used in many other devices by many cultures.

The system was abolished in 1806. A large variety of devices have been invented to measure time.

The study of these devices is called horology. They can be driven by a variety of means, including gravity, springs, and various forms of electrical power, and regulated by a variety of means. A sundial is any device that uses the direction

of sunlight to cast shadows from a gnomon onto a set of markings calibrated to indicate the local time, usually to the hour.

The idea to separate the day into smaller parts is credited to Egyptians because of their sundials, which operated on a duodecimal system. The importance of the number 12 is due to the number of lunar cycles in a year and the number of stars used to count the passage of night. Obelisks made as a gnomon were built as early as c. 3500 BC. An Egyptian device that dates to c.

Henri Bergson believed that time was neither a real homogeneous medium nor a mental construct, but possesses what he referred to as Duration. Duration, in Bergson's view, was creativity and memory as an essential component of reality. According to Martin Heidegger we do not exist inside time, we are time. Hence, the relationship to the past is a present awareness of having been, which allows the past to exist in the present.

Different observers may calculate different distances and different time intervals between two events, but the invariant interval between the events is independent of the observer and their velocity. Unlike space, where an object can travel in the opposite directions (and in 3 dimensions), time appears to have only one dimension and only one direction—the past lies behind, fixed and immutable, while the future lies ahead and is not necessarily fixed.

According to the special theory of relativity, in the high-speed particle's frame of reference, it exists, on the average, for a standard amount of time known as its mean lifetime, and the distance it travels in that time is zero, because its velocity is zero. Relative to a frame of reference at rest, time seems to "slow down" for the particle. Relative to the high-speed particle, distances seem to shorten.

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ABOUT THE AUTHOR

Aira Astralis is a science writer specializing in astronomy and space exploration. Her work spans the full breadth of humanity's relationship with the cosmos -- from the earliest naked-eye observations to the engineering marvels of modern space missions.

Her books are built on a foundation of primary sources: peer-reviewed research, mission archives, and direct engagement with the scientific community. She writes for readers who want to understand not just what we know about the universe, but how we came to know it.

She believes that the most important frontier is not out there but in the gap between what scientists discover and what the public understands. Her books exist to close that gap.